

3.5 PHYLUM-NEMATODA

(*Aschelminthes*, as used in NCERT book)

- Body cavity, a 'one way gut' and 'tube with-in-a tube'

3.5.1 General Characters

The body of a nematode is circular in cross-section, hence the name **s** (Figure 3.9). They may be free living or parasitic on plants and animals, aquatic or terrestrial. Roundworms have organ-system level of organisation. They are bilaterally symmetrical, triploblastic and **pseudocoelomate** animals. Body is unsegmented and covered by a transparent, tough and protective **collagenous cuticle** (a *unique character*). In some, the epidermis is **syncytial**. Alimentary canal is complete with a mouth, well developed muscular pharynx and anus. Excretory system consists of **renette gland** (excretory gland) and ducts. Nervous system consists of a circum enteric nerve ring with ganglia and nerves. Sense organs like **amphids** and **phasmids** occur. Sexes are separate (dioecious). They exhibit sexual dimorphism, often females are longer and larger than males. Males have curved posterior end with a cloacal aperture and one or two copulatory (penial) spicules. Fertilisation is internal, majority are oviparous (e.g. *Ascaris*), a few are ovo-viviparous (e.g. *Wuchereria*) and development may be direct (the young ones resemble the adult- called **juveniles**) or indirect. Growth into adult forms involves typically four **moultings** in the larval or juvenile stages.

Examples: *Ascaris*, *Wuchereria*, *Ancylostoma*.

In nematodes, the number of body cells (except the cells in gonads) remains constant, once the animal reaches the adulthood. It means the number of nuclei too is constant. This phenomenon is called 'eutely'. If there is growth in the size of the body, it is due to the growth in the size of the cells (hypertrophy).

3.5.2 Classification of the phylum Nematoda

Phylum Nematoda is classified into two classes, namely **1. Aphasmidia** and **2. Phasmidia**.

Class-1: Aphasmidia

In the members of this class **amphids** (cuticular depressions around oral region performing chemoreceptor function) are highly modified, but phasmids (posterior glandulo-sensory structures) are absent.

Examples: *Trichinella* (trichinaworm), *Trichuris* (whipworm).

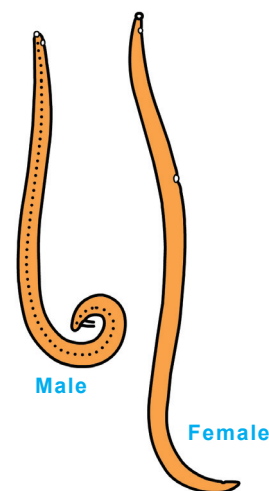


Fig. 3.9 *Ascaris lumbricoides*